ML

School Dropout Prediction using Machine Learning

Python | Scikit-learn | SMOTE | Educational Analytics

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Project Case Study | 2026

The Challenge

Academic failure in first-semester Mathematics can be an early warning signal for student retention and dropout.

The project used historical academic records from IFCE - Fortaleza Campus to estimate whether a student would pass or fail Mathematics in the first semester, enabling earlier pedagogical and managerial interventions.

- ✓ Convert historical records into a supervised classification problem.

Early Risk Detection

RISK

Predict approval/failure probabilities before academic problems become harder to reverse.

Data, Scope and Responsible Use

Dataset

- 468 complete student records used for modeling.
- Six integrated technical programs.
- Entry cohorts from 2018.2 to 2020.1.
- Predictors included admission score, program, admission category and semester.
- Target: pass (grade $\geq$ 6.0) or fail (grade $<$ 6.0) in first-semester Mathematics.

Responsible use: personal identifiers were removed and the modeling used anonymized academic variables.

Dataset at a Glance

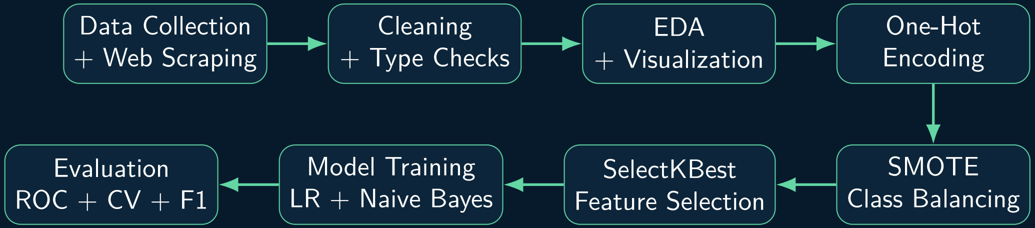
468
complete student records

6
technical programs

2
prediction classes: pass or fail

Source: Pereira et al. (2025), Research, Society and Development, DOI: 10.33448/rsd-v14i6.49029

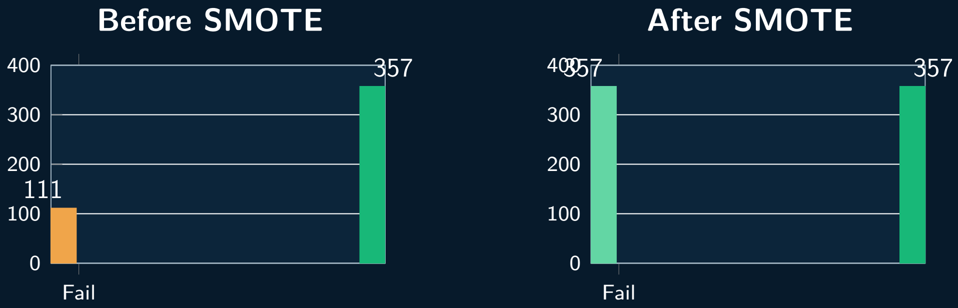
End-to-End Machine Learning Workflow



Python, Pandas, NumPy   Scikit-learn, Matplotlib   SMOTE, Cross-Validation

Source: Pereira et al. (2025), Research, Society and Development, DOI: 10.33448/rsd-v14i6.49029

Handling Class Imbalance



Why this matters

Without balancing, a classifier could achieve deceptively high overall accuracy by favoring the majority class. SMOTE was used to strengthen the model's ability to recognize the minority class: students at risk of failure.

Source: Pereira et al. (2025), Research, Society and Development, DOI: 10.33448/rsd-v14i6.49029

Feature Selection and Model Design

Highest SelectKBest scores

Feature	Score
Admission score	124.20
Chemistry program	30.69
Electrotechnics program	13.20
Admission category L6	10.69
Industrial Mechanics program	10.69
Telecommunications program	10.08

A hybrid filter-wrapper strategy evaluated multiple

score thresholds and selected a cutoff around 10 to balance model simplicity and predictive performance.

Models Compared

- Logistic Regression
- Naive Bayes

Evaluation

- Precision and Recall
- F1-score
- Accuracy
- Cross-validation
- ROC-AUC

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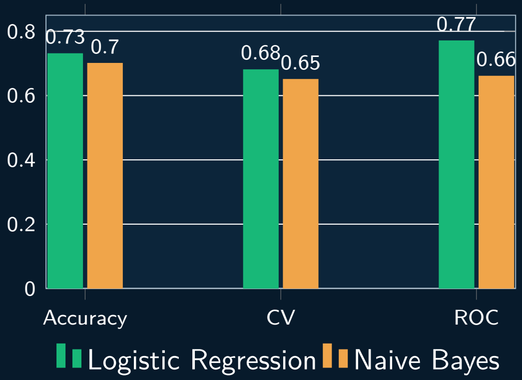
Model Performance



Best Overall Model

Logistic Regression

Higher overall accuracy, cross-validation stability and discriminative performance.

- Accuracy: **0.73**
- Recall for at-risk class: **0.79**
- Cross-validation mean: **0.68**
- ROC-AUC: **0.77**

These results support the use of Logistic Regression as an

Practical Value and Limitations

Potential Institutional Value

- Prioritize students for academic support.
- Inform tutoring and mentoring strategies.
- Produce transparent risk reports for course coordinators.
- Support evidence-based retention policies.

Limitations Reported in the Study

- 468 records from one institution and a limited time window.
- Missing socioeconomic variables.
- Possible dependence among students from the same program/semester.
- Only two classical models were evaluated.

Recommended Next Steps

Expand the dataset, include richer socioeconomic and engagement variables, evaluate additional algorithms, monitor fairness across student groups and validate the model prospectively before operational use.

Peer-Reviewed Publication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Machine learning: Uma aplicação para a prevenção da Evasão Escolar

Machine learning: An application for School Dropout prevention

Aprendizaje automático: Una aplicación para la prevención del Abandono Escolar

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Resumo
A retenção de estudantes em disciplinas de matemática perpetua-se como uma preocupação preeminente nas instituições educacionais, constituindo um desafio multifacetado, pois influencia diretamente a eficácia do processo pedagógico e, consequentemente, incita à evasão escolar. O objetivo do presente estudo é apresentar uma aplicação de aprendizagem de máquina na prevenção de evasão escolar. No escopo desta pesquisa, o intento consiste em discernir as probabilidades associadas à aprovação ou reprovação de alunos matriculados em disciplinas de matemática no primeiro semestre letivo, com base em registros históricos de desempenho acadêmico no Instituto Federal do Ceará (IFCE), campus de Fortaleza. A esse propósito, foi aplicado algoritmos de Aprendizagem de Máquina para indicar a probabilidade de um aluno ser aprovado ou reprovado em matemática no primeiro semestre letivo, gerando relatórios que auxiliem a tomada de decisão por parte das coordenações dos cursos em que essa matéria é lecionada. Serão utilizados os algoritmos Regressão Logística e Naive Bayes. Para isso, serão utilizadas bibliotecas disponíveis para a linguagem Python, a saber, Scikit-learn. O resultado esperado é de que os algoritmos de Aprendizagem de Máquina forneçam percepções na tomada de ações pedagógicas e gerenciais acadêmicas preventivas que visem à redução dos índices de reprovação e, consequentemente, a evasão escolar.
Palavras-chave: Retenção; Evasão; Aprendizado de máquina; *Naive bayes*; Regressão logística.

Abstract
The retention of students in mathematics subjects continues to be a major concern for educational institutions and is a multifaceted challenge, since it directly influences the effectiveness of the teaching process and, consequently, encourages school dropouts. The aim of this study is to present an application of machine learning to prevent school drop-outs. The aim of this research is to discern the probabilities associated with passing or failing students enrolled in mathematics subjects in the first semester of school, based on historical records of academic performance at the Federal Institute of Ceará (IFCE), Fortaleza campus. To this end, Machine Learning algorithms were applied to indicate the probability of a student passing or failing mathematics in the first semester, generating reports to help decision making by the coordinators of the courses in which this subject is taught. The Logistic Regression and Naive Bayes algorithms will be used. To do this, libraries available for the Python language will be used, namely Scikit-learn. The expected result is that the Machine Learning algorithms will provide insights into preventive academic pedagogical and managerial actions aimed at reducing failure rates and, consequently, school dropout.
Keywords: Retention; Dropout; Machine learning; Naive bayes; Logistic regression.

Resumen
La retención de alumnos en las asignaturas de matemática continúa siendo una gran preocupación para las instituciones de enseñanza y constituye un desafío multifacético, ya que influye directamente en la eficacia del proceso pedagógico y, consecuentemente, incita a la deserción escolar. El objetivo de esta investigación es discernir las probabilidades asociadas a la aprobación o reprobación de alumnos matriculados en asignaturas de matemáticas en

Machine Learning: An Application for School Dropout Prevention

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Project Case Study

Based on Pereira et al. (2025)